
Air Quality Prediction in Houston: A Comparative Analysis with Data Leakage Detection

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Abstract

Air pollution poses a major threat to public health in Houston, Texas, which consistently ranks among the worst U.S. cities for ozone pollution due to its industrial corridor and heavy traffic. We investigate the predictability of Houston’s Air Quality Index (AQI) at horizons from 1 to 48 hours using EPA ground station measurements and weather data from Open-Meteo. We compare 10 models spanning tree-based methods (Random Forest, XGBoost, Gradient Boosting), ensemble approaches (stacked meta-learner), time series decomposition (STL-XGBoost), and deep learning (LSTM, hybrid LSTM-GRU). During evaluation, we discover significant data leakage: same-day pollutant readings used as features are direct inputs to the EPA’s AQI formula, inflating Random Forest importance for Ozone_max to 80% and producing artificially strong results ($R^2 = 0.87$). After removing leaky features, we find that daily AQI forecasting is genuinely difficult ($R^2 = 0.29$), while switching to hourly data (24,071 samples) recovers strong performance ($R^2 = 0.855$, MAE = 4.25) driven primarily by temporal autocorrelation. Tree-based models consistently outperform deep learning on this structured tabular data. We complement our forecasting analysis with an environmental justice assessment using EJScreen census tract data, revealing that pollution burden concentrates in lower-income East Houston communities near the Ship Channel industrial corridor. Our findings highlight the importance of proper feature validation in AQI forecasting and demonstrate that data scale matters more than model complexity for this task.

1 Introduction

Houston, Texas is the fourth largest city in the United States and home to one of the world’s largest petrochemical complexes along the Houston Ship Channel. Harris County consistently exceeds EPA air quality standards for ozone and fine particulate matter (PM_{2.5}), making AQI prediction a practical public health concern (1). Accurate short-term forecasts could enable vulnerable populations to take precautionary measures and support policy decisions around industrial operations.

Machine learning approaches for AQI forecasting have proliferated in recent years, with studies reporting strong results using tree-based models (6), recurrent neural networks (7), and hybrid architectures (2). However, a recent review by Ma et al. (4) warns that many AQI prediction studies suffer from methodological issues including data leakage, improper validation splits, and overreliance on same-day features that are components of the target variable.

In this work, we conduct a systematic comparison of 10 models for Houston AQI prediction and make three contributions: (1) we identify and quantify a data leakage problem where same-day pollutant readings inflate model performance, (2) we show that temporal resolution (hourly vs. daily) has a

*Contributions: Sadhana – EDA, feature engineering, report writing (25%). Abdelrahman – modeling, code integration, leakage analysis (25%). Tien – enrichment data collection, EJScreen map, report writing (25%). Tabriz – data pipeline, hourly data processing, advanced modeling, presentation, visualization (25%).

larger impact on forecasting accuracy than model architecture, and (3) we provide an environmental justice analysis showing the spatial distribution of pollution burden across Houston census tracts.

2 Background

Time series decomposition for AQI. Ma et al. (2) proposed combining STL (Seasonal-Trend decomposition using Loess) with deep learning and intelligent optimization for AQI prediction, demonstrating that decomposing the signal into trend, seasonal, and residual components improves forecasting accuracy. We adopt STL decomposition as one of our modeling approaches, combined with XGBoost rather than deep learning given our dataset size.

Optimized feature selection. Natarajan et al. (3) used Grey Wolf Optimization with Decision Trees for AQI prediction across Indian cities, motivating our use of Optuna for automated hyperparameter search. Their work achieved R^2 values of 0.88–0.97 depending on city, though Indian cities exhibit greater AQI variance than Houston.

Interpretable ML for air quality. Houdou et al. (5) conducted a systematic review finding SHAP as the most commonly used interpretability method for AQI models. We use SHAP analysis both for model interpretation and, critically, for diagnosing data leakage.

Methodological pitfalls. Ma et al. (4) reviewed overlooked issues in air quality ML, specifically warning about feature leakage when current pollutant concentrations are used to predict AQI. Our work independently discovered this exact problem and provides empirical evidence of its severity.

3 Data

3.1 Sources

We combine data from multiple publicly available sources. **EPA Air Quality System (AQS):** Hourly measurements of ozone, PM_{2.5}, SO₂, CO, and NO₂ from ground monitoring stations in Harris County, spanning January 2023 to September 2025 (24,071 hourly records after preprocessing). We also collected daily AQI summaries from 2022–2025 (1,340 daily records). **Open-Meteo:** Hourly weather variables including temperature, relative humidity, wind speed and direction, precipitation, pressure, cloud cover, and shortwave radiation, matched to EPA timestamps. **Enrichment sources:** EPA Toxic Release Inventory (TRI) facility-level emissions data (6,519 records), EPA Facility Registry Service (FRS) locations, and EJScreen 2024 census tract pollution indicators (1,562 Houston-area tracts) archived from Zenodo after EPA took the tool offline in February 2025.

3.2 Preprocessing

Missing values from sensor downtime were handled using forward-fill followed by backward-fill, appropriate for time series where the most recent available reading is the best estimate. We note that PM_{2.5} hourly data for early 2023 had substantial missingness (~36%) due to column format inconsistencies across EPA file versions; these gaps were interpolated. After cleaning, we compute an approximate AQI from pollutant sub-indices following EPA breakpoint tables (1).

3.3 Feature Engineering

We engineer 64 features across six categories: (1) **Lag features** – AQI at 1, 2, 3, 6, 12, and 24 hours prior, plus pollutant lags for ozone, PM_{2.5}, and NO₂; (2) **Rolling statistics** – mean and standard deviation over 3, 6, 12, and 24-hour windows; (3) **Rate of change** – AQI differences over 1, 6, and 24-hour intervals; (4) **Cyclical encoding** – sine/cosine transforms of hour, day-of-week, and month; (5) **Time indicators** – hour, day of week, weekend flag; (6) **STL components** – trend, seasonal, and residual from Seasonal-Trend decomposition with period 7.

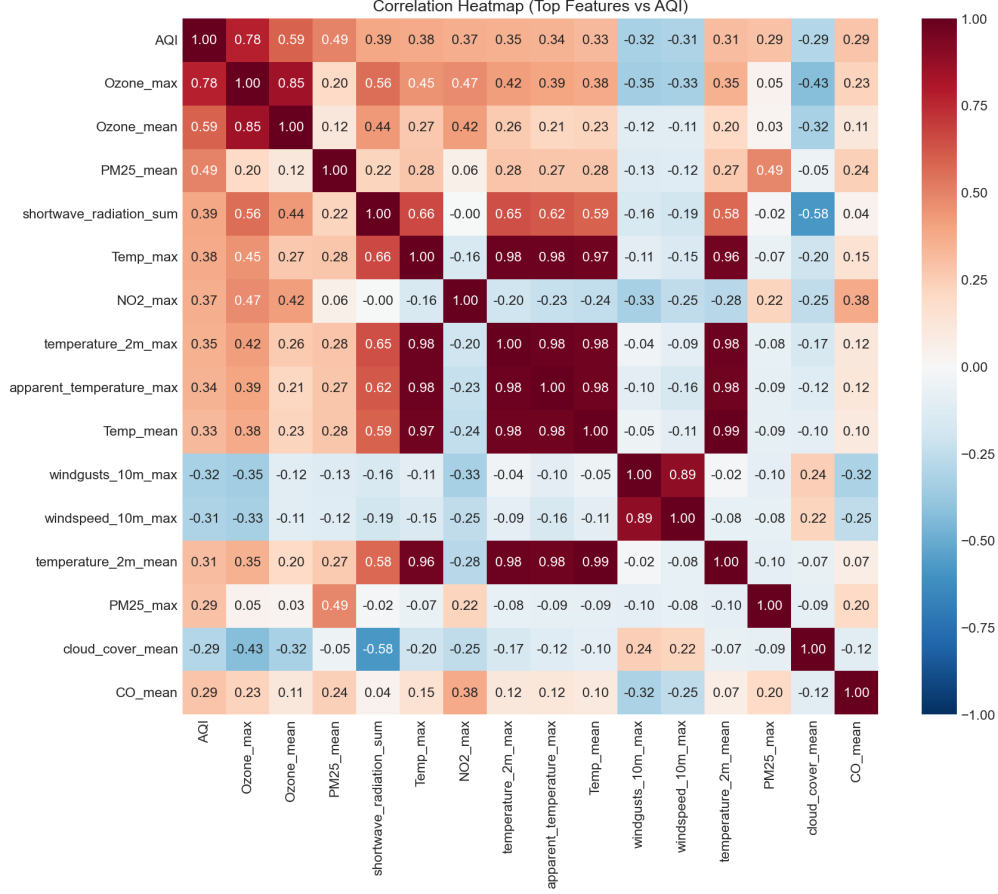


Figure 1: Correlation heatmap of top features vs. AQI. Ozone_max (0.78) and Ozone_mean (0.59) show the strongest correlations, followed by PM2.5 and temperature variables.

4 Methods

All models use a time-based 80/20 train-test split to prevent temporal leakage. We evaluate both regression metrics (MAE, RMSE, R^2 , MAPE) and AQI category classification (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy).

Persistence baseline. Predicts next hour’s AQI equals current hour’s AQI. Any useful model must beat this.

Tree-based models. Random Forest (200 trees, max depth 20) and Gradient Boosting (300 estimators, depth 5, learning rate 0.05) from scikit-learn, plus XGBoost (6) with hyperparameters optimized via Optuna (50 Bayesian trials with 3-fold time series cross-validation). We experimented with Optuna tuning on all three models but found that XGBoost benefited most from tuning due to its larger hyperparameter search space. Random Forest and Gradient Boosting showed minimal improvement over manual defaults, so we kept the manually tuned versions for clarity.

Stacked ensemble. Out-of-fold predictions from XGBoost, Random Forest, and Gradient Boosting are combined via a Ridge regression meta-learner ($\alpha = 1.0$).

STL-XGBoost hybrid. Inspired by (2), we decompose AQI into trend, seasonal, and residual components, train separate XGBoost models on each, and sum the predictions.

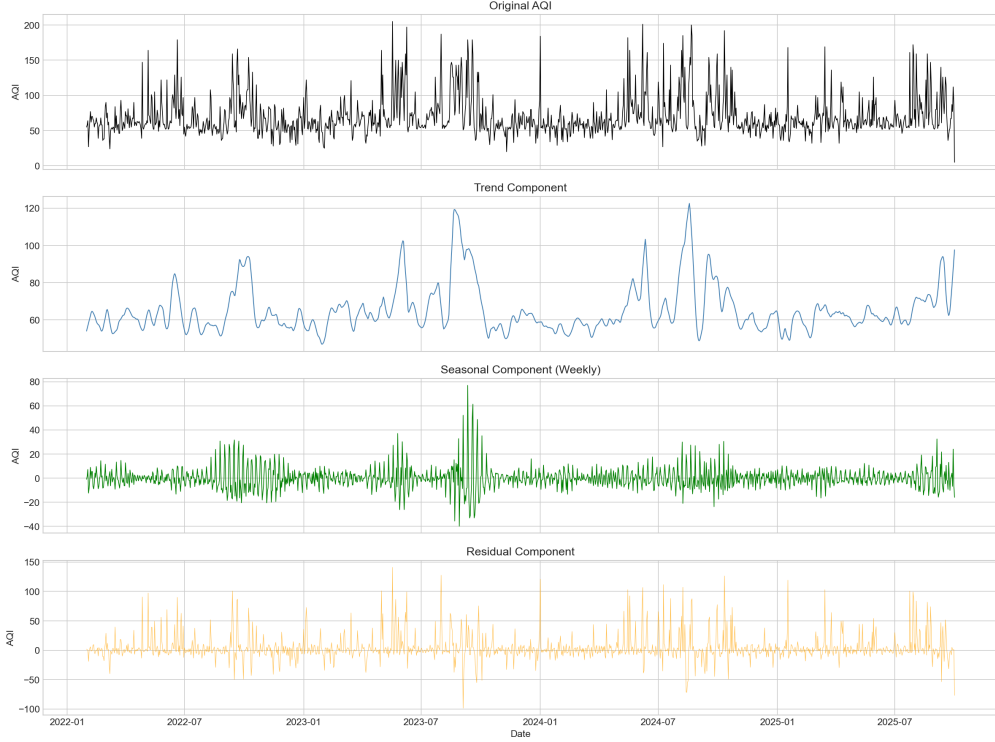


Figure 2: STL decomposition of Houston AQI (2022–2025). Top to bottom: original signal, trend component, weekly seasonal pattern, and residual. Clear seasonal structure motivates the decomposition approach.

Deep learning. An LSTM (3 layers, 512 units) and a hybrid LSTM-GRU with dual branches (64-unit LSTM + 64-unit GRU concatenated, trained with Huber loss and learning rate scheduling). Lookback window of 14 timesteps with MinMax scaling.

5 Experiments

5.1 Data Leakage Discovery

Our initial experiments on daily data with all features produced apparently strong results (Table 4). However, feature importance analysis revealed that `Ozone_max` alone accounted for 80.8% of Random Forest’s decisions (Figure 3). Since AQI is defined as the maximum of pollutant sub-indices, and ozone is the dominant pollutant in Houston, using same-day `Ozone_max` to predict AQI is mathematically circular. SHAP analysis confirmed that the top contributors were exclusively same-day pollutant readings (5; 4).

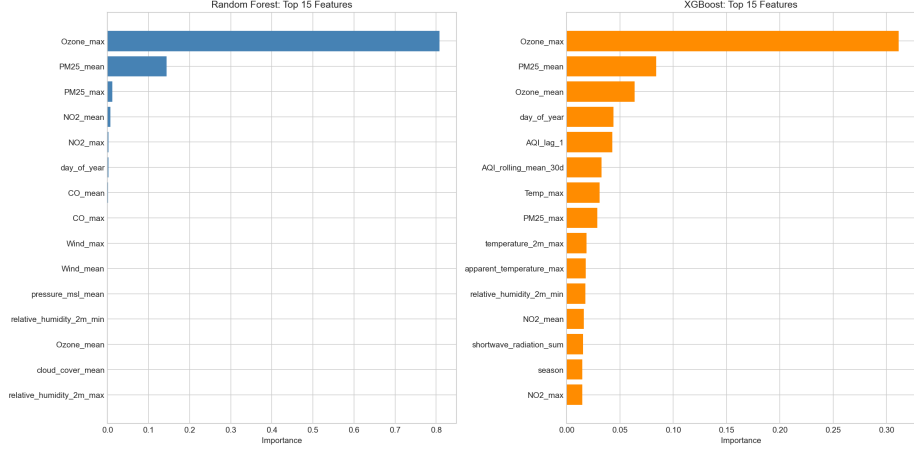


Figure 3: Feature importance from Random Forest (left) and XGBoost (right) with all features. **Ozone_max** dominates RF at 80.8%, revealing data leakage since AQI is computed from pollutant sub-indices.

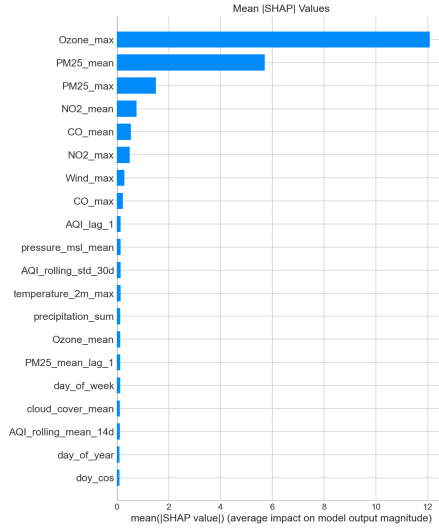


Figure 4: SHAP mean absolute values confirm the leakage: **Ozone_max** and **PM25_mean** are the top two contributors, both of which are same-day pollutant readings used in the AQI calculation.

To quantify the leakage, we compare models trained with all features (“leaky”) versus only features available at prediction time (“fair”) on hourly data:

Table 1: Leakage quantification on hourly data. The 6–16 $\times$ MAE difference between leaky and fair features confirms that same-day pollutants are the primary source of predictive power in the leaky setting.

Model	Leaky MAE	Fair MAE	Ratio
Random Forest	0.27	4.37	16 $\times$
XGBoost	0.68	4.26	6 $\times$

5.2 Fair-Feature Results: Daily vs. Hourly

After removing 10 leaky same-day pollutant features, we retrain all models using only fair features (lagged values, weather, time features). On daily data (1,340 samples), the best model achieves $R^2 = 0.29$ (Table 2), indicating that genuine daily AQI forecasting with limited data is difficult.

Table 2: Fair-feature results on daily data (1,340 samples). Performance drops substantially without leaky features, revealing the true difficulty of AQI forecasting.

Model	MAE	RMSE	R^2
Gradient Boosting	14.25	21.00	0.290
XGBoost (Optuna)	14.39	21.35	0.266
Random Forest	14.50	21.37	0.264
Persistence (yesterday)	17.88	28.82	-0.338

Switching to hourly data (24,071 samples) dramatically improves results (Table 3). Gradient Boosting achieves $\text{MAE} = 4.25$ and $R^2 = 0.855$. However, the persistence baseline is already strong ($R^2 = 0.812$) because hourly AQI is highly autocorrelated. The improvement over persistence is genuine but modest: our best model reduces MAE by 11% (from 4.77 to 4.25).

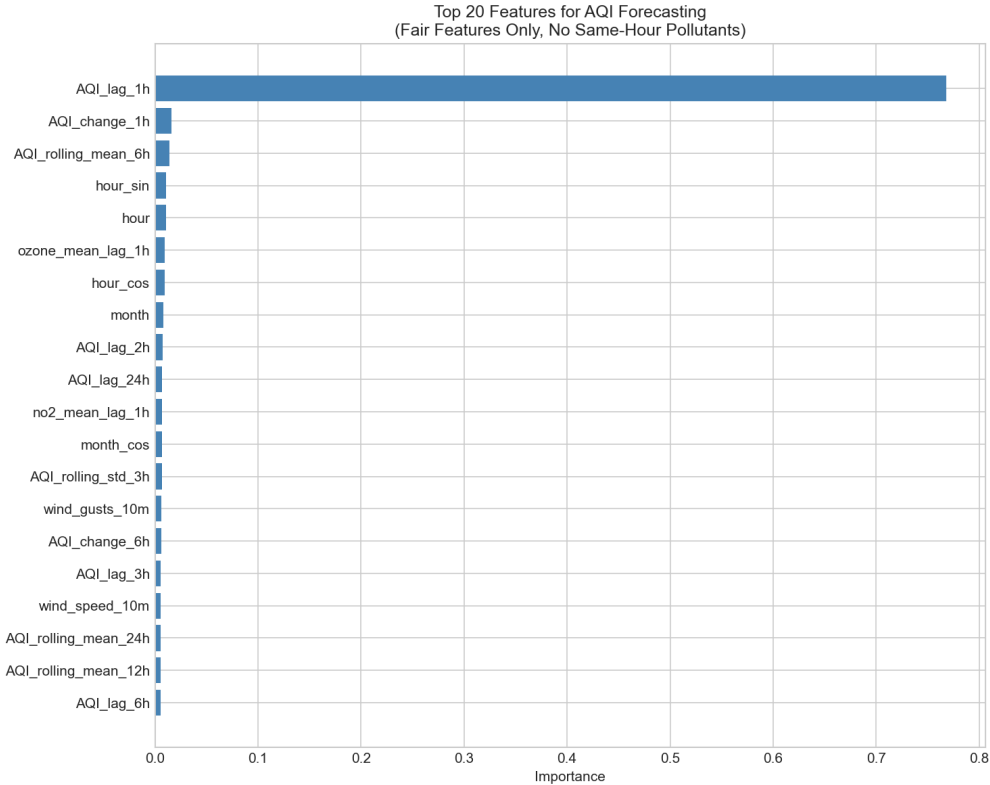


Figure 5: Feature importance with fair features only (hourly). `AQI_lag_1h` dominates at 77%, confirming the model primarily leverages temporal autocorrelation. Time-of-day and rolling statistics provide marginal additional signal.

Table 3: Fair-feature results on hourly data (24,071 samples). All models beat persistence, with tree-based methods outperforming deep learning.

Model	MAE	RMSE	R^2	MAPE
Gradient Boosting	4.25	6.35	0.855	10.2%
XGBoost (Optuna)	4.26	6.37	0.855	10.2%
Random Forest	4.37	6.50	0.849	10.4%
Linear Regression	4.70	6.90	0.829	11.2%
Persistence	4.77	7.24	0.812	11.1%
Stacked Ensemble	4.84	6.98	0.825	11.2%

Feature importance on the fair hourly model reveals that `AQI_lag_1h` accounts for 77% of XGBoost’s importance. Time-of-day features (`hour_sin`, `hour_cos`), rolling statistics, and ozone/ NO_2 lags contribute the remaining signal. This confirms that the model primarily leverages temporal autocorrelation, with weather and diurnal patterns providing marginal additional value.

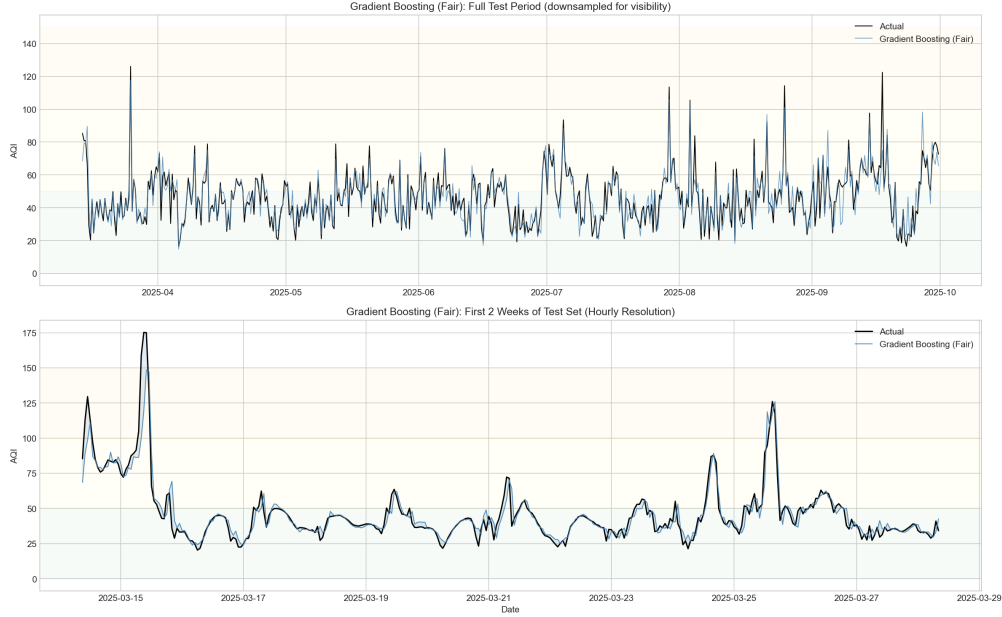


Figure 6: Predicted vs. actual AQI for Gradient Boosting (fair features, hourly). Top: full test period. Bottom: 2-week zoom showing the model tracks daily cycles but undershoots spikes.

5.3 Classification Performance

For AQI category classification, the hourly fair-feature model achieves 88% overall accuracy. Good (91% precision) and Moderate (76% precision) categories are well-predicted, but Unhealthy events have very low recall due to severe class imbalance (only 5 Unhealthy observations in the test set). On daily data with fair features, classification accuracy drops to 81% and Unhealthy days have 0% recall, highlighting the practical limitation that the most safety-critical predictions are the least reliable.

5.4 Prediction Horizon Analysis

AQI predictability degrades rapidly with horizon: 1 hour ahead yields $R^2 = 0.855$ (hourly fair), 24 hours ahead yields $R^2 = 0.29$ (daily fair), and 48 hours ahead yields $R^2 = -0.19$ (worse than predicting the mean). This is consistent with the autocorrelation structure: lag-1 hourly AQI correlation is $r > 0.95$, but this signal decays as the gap between known information and prediction target widens.

The poor performance of the 48-hour forecasting models may be due to the dominance of short-term temporal features, especially `AQI_lag_1h`, which contributed about 77% of the model's predictive power. As the prediction horizon increased, the autocorrelation signal weakened, causing model performance to decline. This suggests the models relied more on immediate past observations than long-range temporal patterns.

5.5 Environmental Justice Analysis

Using EJScreen 2024 census tract data overlaid with TRI facility locations and EPA monitoring stations (Figure 7), we observe that PM_{2.5} concentrations and toxic release volumes are significantly higher in East Houston census tracts near the Ship Channel industrial corridor. These areas overlap with lower-income and minority communities, confirming the environmental justice dimension of air quality in Houston. Accurate AQI forecasting has direct public health value for these disproportionately affected communities.

We also note that monitoring station density is lower in East Houston's most pollution-burdened tracts, meaning model predictions in these areas rely on interpolated readings rather than direct measurements.

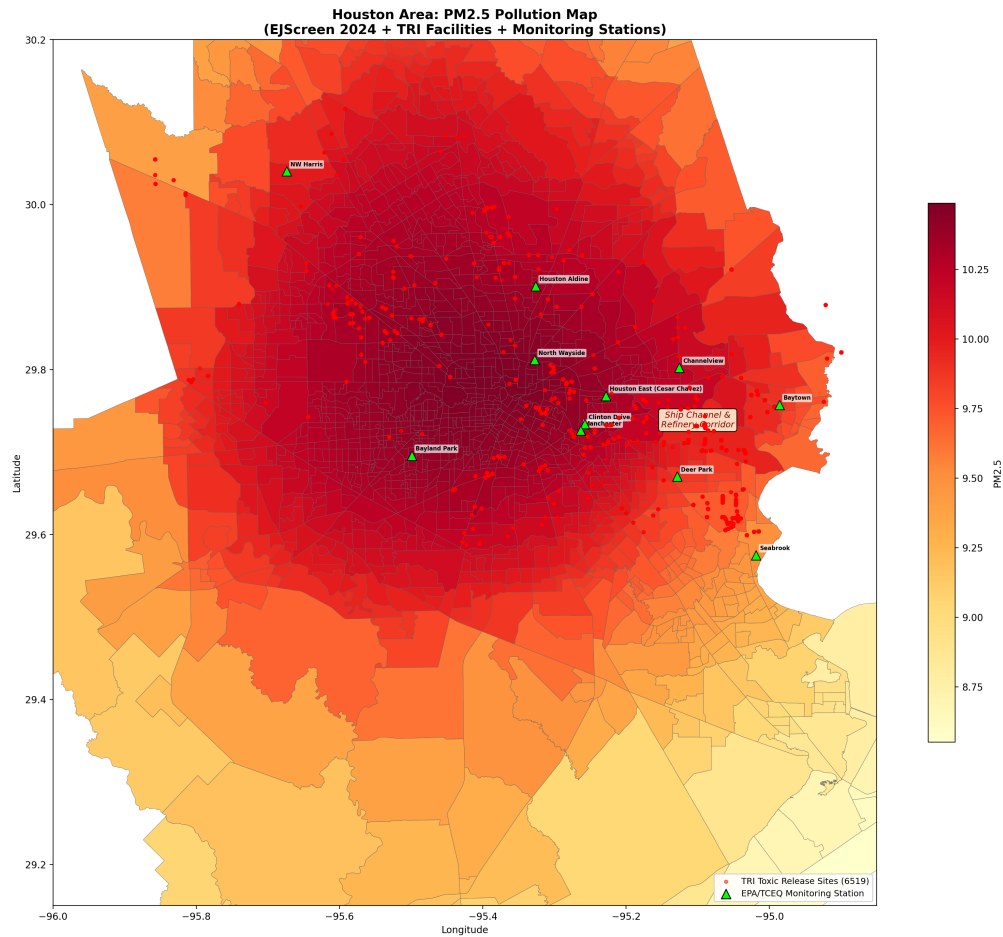


Figure 7: Houston PM_{2.5} pollution map. Census tracts colored by EJScreen PM_{2.5} concentration, with TRI toxic release sites (red) and EPA/TCEQ monitoring stations (green triangles). Pollution concentrates near the Ship Channel in East Houston.

5.6 Leaky-Feature Results (for Reference)

Table 4: Results with all features including same-day pollutants (leaky). These represent AQI *reconstruction* from component readings, not genuine forecasting.

Model	MAE	RMSE	R^2	MAPE
Stacked Ensemble	3.33	9.29	0.861	10.3%
XGBoost (Optuna)	3.36	8.88	0.873	10.5%
STL-XGBoost	4.68	9.65	0.850	12.4%
Hybrid LSTM-GRU	15.41	22.84	0.160	28.5%

6 Conclusion

We conducted a systematic comparison of machine learning models for Houston AQI prediction and identified three key findings. First, **data leakage from same-day pollutant features** is a serious risk in AQI forecasting that inflates performance metrics from $R^2 = 0.85$ to $R^2 = 0.99$; SHAP analysis proved effective for diagnosing this issue.

Second, **temporal resolution matters more than model architecture**: switching from daily (1,340 samples, $R^2 = 0.29$) to hourly data (24,071 samples, $R^2 = 0.855$) had a far larger impact than any modeling improvement.

Third, **tree-based models consistently outperform deep learning** on this structured tabular data, with LSTM achieving $R^2 = 0.16$ compared to Gradient Boosting’s $R^2 = 0.855$ on fair features. The LSTM’s poor performance likely reflects insufficient data volume, deep sequence models typically require far more than 24,071 samples to outperform tree-based methods on tabular data.

We note an important caveat: the hourly $R^2 = 0.855$ is largely driven by temporal autocorrelation (the 1-hour lag accounts for 77% of feature importance), and the persistence baseline already achieves $R^2 = 0.812$. The genuine forecasting challenge lies in daily and multi-day prediction, where performance is substantially weaker. High-AQI events remain difficult to predict due to class imbalance, which is unfortunate since these are the most consequential for public health.

Future work could integrate traffic count data (TxDOT AADT), develop per-station spatial models instead of county aggregates, explore multi-horizon forecasting with separate models for 1h, 6h, and 24h prediction, investigate oversampling or cost-sensitive learning to improve recall on rare high-AQI events, and explore longer lookback windows, transformer-based architectures, additional meteorological features, and multi-step forecasting strategies for long-term AQI prediction.

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